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7、

令 $u=v+w$, $v=\frac{A}{l}x\sin\omega t$

$$\Rightarrow \begin{cases} \frac{dw}{dt}-k\frac{d^2w}{dx^2}+\frac{A}{l}x\cos\omega t=0 \\ w|_{x=0}=0,\quad w|_{x=l}=A\sin\omega t \\ w|_{t=0}=0 \end{cases}$$

$$-\frac{8A\pi^2}{(4n+1)^2}\int_0^t e^{-k(\quad)^2 t}\sin\omega t\,dt$$

$$\Rightarrow \text{令 } w=\sum T(t)\sin\frac{2n+1}{2l}\pi x$$

$$\Rightarrow T'+k\left(\frac{4n+1}{2l}\pi\right)^2 T+\frac{8A\pi^2}{(4n+1)^2}\sin\omega t=0$$

$$\int=-e\cos-k(\quad)^2 e\sin -k(\quad)^2 dt$$

解得: $T(t)=\frac{8A\pi^2}{\omega(4n+1)^2}\cdot\frac{\cos\omega t+k(\frac{4n+1}{2l}\pi)^2\sin\omega t}{1+k^2(\frac{4n+1}{2l}\pi)^4}$

$$\Rightarrow w=\sum\frac{8A\pi^2}{\omega(4n+1)^2}\cdot\frac{\cos\omega t+k(\frac{4n+1}{2l}\pi)^2\sin\omega t}{1+k^2(\frac{4n+1}{2l}\pi)^4}\sin\frac{2n+1}{2l}\pi x$$

$$-\frac{\omega+k(\quad)^2\sin}{1+k^2(\quad)^4}e$$

$u=v+w$

$$=\sum_{n=0}^{\infty}\frac{8A\pi^2}{\omega(4n+1)^2}\cdot\frac{\cos\omega t+k(\frac{4n+1}{2l}\pi)^2\sin\omega t}{1+k^2(\frac{4n+1}{2l}\pi)^4}\sin\frac{2n+1}{2l}\pi x+\frac{A}{l}x\sin\omega t$$

12、令 $u=v+w$

令 $v=A\cos\omega t$, 即:

$$\begin{cases} \frac{dw}{dt}-k\frac{d^2w}{dx^2}=\omega A\sin\omega t \\ w|_{x=0}=0,\quad w|_{x\to\infty}\text{有界} \end{cases}$$

$$\Rightarrow w=\sum_{n=0}^{\infty}\frac{\frac{1-(-1)^n}{n}\frac{4A}{\pi}\omega}{\omega^2+k^2\frac{n^2\pi^2}{l^2}}\left[\sin\omega t-\frac{\omega l^2}{kn^2\pi^2}\cos\omega t\right]\sin\frac{n\pi}{l}x$$

当 $l\to\infty$ 时,

$$w=\int_0^{\infty}\frac{4A\omega}{\pi k^2\alpha}\frac{\sin\alpha x}{\frac{\omega^2}{k^2}+\alpha^4}\left[\sin\omega t-\frac{\omega}{k}\frac{1}{\alpha^2}\cos\omega t\right]d\alpha$$

然后积分不会积了...

A.4

1. 由题意，列出方程，

$$\begin{cases} \nabla^2 u = 0 \Rightarrow \frac{1}{r}\frac{d}{dr}\left(r\frac{du}{dr}\right) + \frac{1}{r^2}\frac{d^2u}{d\varphi^2} = 0 \\ u|_{\varphi, r=a} = V, \quad 0 < \varphi < \pi \\ u|_{\varphi, r=a} = -V, \quad \pi < \varphi < 2\pi \\ u|_{\varphi=2\pi} = u|_{\varphi=0}, \quad \frac{du}{d\varphi}\Big|_{\varphi=2\pi} = \frac{du}{d\varphi}\Big|_{\varphi=0} \\ u|_{r=0} \text{有界} \end{cases}$$

令 $u = R(r)\Phi(\varphi)$，代入，

$$\begin{cases} \Phi'' - \lambda\Phi = 0 \\ R'' + R' - \frac{r^2}{\lambda}R = 0 \\ \Phi_{(0)} = \Phi_{(2\pi)}, \quad \frac{d}{d\varphi}\Phi_{(0)} = \frac{d}{d\varphi}\Phi_{(2\pi)} \end{cases}$$

得：

$\lambda = 0$ 时，$\Phi_{0(\varphi)} = 1$

$\lambda = m^2$ 时，$\Phi_{m(\varphi)} = \sin m\varphi, \cos m\varphi$

又 R 有：$R = r^l$，代入，

$$R_{(r)} = r^m, r^{-m} \quad C_0 + \left[C_{m_1}\sin m\varphi + C_{m_2}\cos m\varphi\right]a^m = V$$

$\lambda = 0$ 时，$R = C_0 + D_0 \ln r$

$\Rightarrow$ 一般解：$u = C_0 + D_0\ln r + \sum_{m=0}^{\infty}\left[\left(C_{m_1}r^m + D_{m_1}r^{-m}\right)\sin m\varphi + \left(C_{m_2}r^m + D_{m_2}r^{-m}\right)\cos m\varphi\right]$

代入边调，

$$u = \frac{V}{2} + \sum_{n=0}^{\infty}\frac{V}{2a^n}\left[\sin m\varphi + \cos m\varphi\right]r^n, \quad 0 < \varphi < \pi$$

$$u=\frac{-V}{2}-\sum_{n=0}^{\infty}\frac{V}{2a^n}\left[\sin n\varphi+\cos n\varphi\right]r^n,\quad \pi<\varphi<2\pi$$

3、由题，

$$\begin{cases}\nabla^2 u=0\\ u|_{r=a}=f(\varphi),\ \dfrac{du}{dr}\Big|_{r=b}=g(\varphi)\\ u|_{\varphi=0}=u|_{\varphi=2\pi},\ \dfrac{du}{d\varphi}\Big|_{\varphi=0}=\dfrac{du}{d\varphi}\Big|_{\varphi=2\pi}\end{cases}$$

一般解为：

$$u=C_0+D_0\ln r+\sum_{m=1}^{\infty}\left[\left(C_{m_1}r^m+D_{m_1}r^{-m}\right)\sin m\varphi+\left(C_{m_2}r^m+D_{m_2}r^{-m}\right)\cos m\varphi\right.$$

代入边值

$$\begin{cases}C_0+D_0\ln a=\dfrac{1}{2\pi}\displaystyle\int_0^{2\pi}f(\varphi)d\varphi\\ D_0=\dfrac{b}{2\pi}\displaystyle\int_0^{2\pi}g(\varphi)d\varphi\\ C_{m_1}a^m+D_{m_1}a^{-m}=\dfrac{1}{\pi}\displaystyle\int_0^{2\pi}f(\varphi)\sin m\varphi d\varphi\\ C_{m_2}a^m+D_{m_2}a^{-m}=\dfrac{1}{\pi}\displaystyle\int_0^{2\pi}f(\varphi)\sin m\varphi d\varphi\\ C_{m_1}b^{m-1}-D_{m_1}b^{-m-1}=\dfrac{1}{m\pi}\displaystyle\int_0^{2\pi}g(\varphi)\sin m\varphi d\varphi\\ C_{m_2}b^{m-1}-D_{m_2}b^{-m-1}=\dfrac{1}{m\pi}\displaystyle\int_0^{2\pi}g(\varphi)\sin m\varphi d\varphi\end{cases}$$

代入，得解

5、

列出方程，

$$\begin{cases} \nabla^2 u = r^3 \sin 2\varphi \\ u|_{r=a} = 0 \\ u|_{\varphi=0} = u|_{\varphi=2\pi},\ \dfrac{\partial u}{\partial \varphi}\Big|_{\varphi=0} = \dfrac{\partial u}{\partial \varphi}\Big|_{\varphi=2\pi} \\ u|_{r=0} \text{有界} \end{cases}$$

特解为：

$$v = \frac{1}{21} r^5 \sin 2\varphi$$

令 $u = v + w$

$$\Rightarrow \begin{cases} \nabla^2 w = 0 \\ w|_{r=a} = -\frac{1}{21} a^5 \sin 2\varphi \\ w|_{\varphi=0} = w|_{\varphi=2\pi},\ \dfrac{\partial u}{\partial \varphi}\Big|_{\varphi=0} = \dfrac{\partial u}{\partial \varphi}\Big|_{\varphi=2\pi} \\ u|_{r=0} \text{有界} \end{cases}$$

$$\Rightarrow \quad w = C_0 + \sum_{m=1}^{\infty} \left[C_m \sin m\varphi + D_m \cos m\varphi \right] r^m$$

代入边调，

$$C_0 = \frac{1}{2\pi} \int_0^{2\pi} -\frac{1}{21} a^5 \sin 2\varphi \, d\varphi$$

$$= 0$$

$$C_m = \frac{1}{\pi} \int_0^{2\pi} -\frac{1}{21} a^5 \sin 2\varphi \sin m\varphi \, d\varphi$$

$$= \begin{cases} 0, & m \neq 2 \\ -\frac{1}{21} a^5, & m = 2 \end{cases}$$

$$D_m = \begin{cases} 0, & m \neq 2 \\ -\frac{1}{21} a^5, & m = 2 \end{cases}$$

$$\Rightarrow W = -\frac{a^3}{v} r^2 \left[\sin\theta + \cos\theta \right]$$

$$u = v + w$$

$$= \frac{1}{v} \left[r^3 \sin\theta - r^3 a \left[\sin\theta + \cos\theta \right] \right]$$